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Causal Discovery Algorithms: Descriptions, Mathematical Formulations, and Causal Inference

1. Linear Non Gaussian Acyclic Model

The Linear Non Gaussian Acyclic Model identifies directed causal relationships from observational data by exploiting linear structural relationships and statistically independent, non Gaussian disturbances. The method assumes that the underlying causal graph is acyclic, that relevant common causes are observed, and that the disturbances are mutually independent. Its use of non Gaussian disturbances helps distinguish causal directions that may not be identifiable from correlations alone.

The structural equation is

$$\mathbf{x} = \mathbf{B}\mathbf{x} + \mathbf{e}$$

where $\mathbf{x}$ is the vector of observed variables, $\mathbf{B}$ is the weighted adjacency matrix, and $\mathbf{e}$ is a vector of independent, non Gaussian disturbances. Under this convention, a nonzero coefficient B_{ij} represents a direct causal relationship from variable X_j to variable X_i.

Rearranging gives

$$\mathbf{x} = (\mathbf{I} - \mathbf{B})^{-1}\mathbf{e}$$

The algorithm estimates the causal ordering and the coefficients by exploiting the statistical independence of the disturbances, commonly through independent component analysis or direct causal ordering. A nonzero estimated coefficient indicates a candidate direct causal relationship, subject to the model assumptions.

Cyclic extension of the Linear Non Gaussian Model

A cyclic extension permits feedback relationships between variables. The structural equation remains

$$\mathbf{x} = \mathbf{B}\mathbf{x} + \mathbf{e}$$

but $\mathbf{B}$ is no longer restricted to representing an acyclic graph. The solution is

$$\mathbf{x} = (\mathbf{I} - \mathbf{B})^{-1}\mathbf{e}$$

provided that $\mathbf{I} - \mathbf{B}$ is invertible. Additional conditions are required to establish stability and identifiability.

This formulation permits cycles such as $X_1 \to X_2 \to X_1$, representing feedback between variables. However, allowing cycles does not automatically guarantee that the individual causal coefficients can be uniquely identified. The assumptions about the disturbances, stability, and the data generating process must remain appropriate. The cyclic extension must be implemented explicitly because standard Linear Non Gaussian Acyclic Model algorithms generally assume an acyclic graph.

2. Granger Causality

Granger causality evaluates whether the historical observations of one variable improve the prediction of another variable beyond the information contained in the target variable's own history. It is designed for time series and can identify delayed relationships and feedback between variables. The method establishes predictive causality rather than proving that an intervention on one variable would necessarily change another.

For a source variable X and a target variable Y, the restricted model is

$$Y_t = \sum_{k=1}^{p} a_k Y_{t-k} + \epsilon_t$$

The unrestricted model incorporates historical values of the source variable:

$$Y_t = \sum_{k=1}^{p} a_k Y_{t-k} + \sum_{k=1}^{p} b_k X_{t-k} + u_t$$

The null hypothesis is

$$H_0 : b_1 = b_2 = \cdots = b_p = 0$$

The null hypothesis states that the historical values of X provide no additional predictive information about Y. For nested linear models, the hypothesis can be evaluated using the following statistic:

$$F = \frac{(RSS_R - RSS_U)/p}{RSS_U/(T - k_U)}$$

Here, RSS_R and RSS_U denote the residual sums of squares from the restricted and unrestricted models, p is the number of tested lag coefficients, T is the number of usable observations, and k_U is the number of parameters in the unrestricted model.

A statistically significant result supports the conclusion that X Granger causes Y at the selected lag order. Appropriate lag selection, stationarity or a suitable model for nonstationary data, and consideration of relevant confounding variables are essential for reliable interpretation.

3. Peter and Clark Algorithm

The Peter and Clark algorithm discovers causal structures through repeated conditional independence tests. It begins with a fully connected undirected graph and removes connections when the corresponding variables are found to be conditionally independent given a suitable subset of other variables. It then uses orientation rules to identify possible causal directions.

The fundamental statistical test is

$$H_0 : X_i \perp\!\!\!\perp X_j \mid \mathbf{S}$$

where X_i and X_j are observed variables and $\mathbf{S}$ is a conditioning set containing selected other variables.

For continuous data under suitable distributional assumptions, conditional independence can be tested using partial correlation. The test statistic is

$$t = \rho_{ij|S} \sqrt{\frac{n - |S| - 2}{1 - \rho_{ij|S}^2}}$$

where $\rho_{ij|S}$ is the partial correlation, n is the sample size, and $|S|$ is the number of conditioning variables.

If the conditional independence hypothesis is not rejected, the algorithm removes the edge between the variables. If an edge remains, orientation rules may determine its direction based on the surrounding graph and the inferred independence relationships.

The method generally identifies a class of observationally equivalent directed acyclic graphs rather than a unique causal graph. Its standard formulation assumes an acyclic causal structure, no unobserved common causes, and appropriate statistical conditions connecting the graph to the observed distribution.

4. Peter and Clark Momentary Conditional Independence

Peter and Clark Momentary Conditional Independence extends constraint based causal discovery to time series by explicitly testing relationships between variables at different time points. The method uses a preliminary parent selection stage followed by momentary conditional independence testing. This approach helps distinguish direct lagged causal effects from associations explained by autocorrelation and other observed variables.

To test whether the historical value $X_{t-\tau}^i$ causes the present value X_t^j, the method evaluates conditional independence given selected parents of both variables:

$$X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \left(\widehat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \widehat{\mathcal{P}}(X_{t-\tau}^i)\right)$$

Here, τ denotes the time lag and $\widehat{\mathcal{P}}$ denotes an estimated parent set. The conditioning set includes the selected parents of the target and the selected parents of the source, excluding the tested source value from the target's parent set.

The conditional independence test can use partial correlation for appropriate linear relationships or a nonlinear dependence test for more general relationships.

If the source and target remain conditionally dependent after accounting for these variables, the method identifies a candidate causal link at lag τ. The statistical significance of the resulting test is used to assess the evidence for that link.

The method is designed to handle autocorrelated time series and sparse causal structures. Its reliability nevertheless depends on suitable conditional independence tests, adequate observations, and appropriate assumptions about the stability of the underlying causal process. The extension known as

Peter and Clark Momentary Conditional Independence Plus also supports the discovery of contemporaneous relationships under its assumptions.

5. Noncombinatorial Optimization via Trace Exponential for Structure Learning

Noncombinatorial Optimization via Trace Exponential for Structure Learning learns a weighted causal graph by expressing structure discovery as a continuous optimization problem. It estimates the weights of candidate causal relationships while penalizing unnecessary connections and enforcing the requirement that the resulting graph contains no directed cycles.

For a linear structural equation model, the objective function is

$$\min_{\mathbf{W}} \frac{1}{2n} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda \|\mathbf{W}\|_1$$

subject to the acyclicity constraint

$$h(\mathbf{W}) = \operatorname{tr}\left(e^{\mathbf{W} \circ \mathbf{W}}\right) - d = 0$$

Here, $\mathbf{X}$ is the data matrix, $\mathbf{W}$ is the weighted adjacency matrix, n is the number of observations, d is the number of variables, and λ controls sparsity. The symbol $\circ$ denotes elementwise multiplication, while tr denotes the matrix trace.

The first term measures prediction error, while the second penalizes excessive connections. The acyclicity constraint ensures that the estimated graph is a directed acyclic graph. Under this matrix convention, a nonzero entry W_{ij} represents a candidate edge from X_i to X_j.

The algorithm estimates causal relationships through the optimized edge weights. A nonzero weight represents a candidate direct relationship under the fitted model, although statistical significance and causal interpretation require separate assessment.

The standard formulation uses a linear structural model and does not permit arbitrary directed cycles. Nonlinear extensions and time series adaptations are available. For example, Dynamic NOTEARS explicitly models temporal relationships and lagged effects.

6. Greedy Equivalence Search

Greedy Equivalence Search discovers causal structures by maximizing a score that measures how well a candidate graph explains the observed data while accounting for model complexity. Instead of searching every possible directed acyclic graph, it searches over equivalence classes of graphs that imply the same conditional independence relationships.

The method begins with an empty graph and repeatedly adds edges that improve the score. It then removes edges when doing so improves the score. The search terminates when no permitted local modification produces a better result.

A commonly used scoring function is the Bayesian Information Criterion expressed in maximization form:

$$S(G; \mathbf{X}) = \log L(\hat{\theta}_G; \mathbf{X}) - \frac{k_G}{2} \log n$$

Here, G denotes the candidate graph, L is the maximized likelihood, $\hat{\theta}_G$ represents the estimated model parameters, k_G is the number of free parameters, and n is the number of observations.

The algorithm evaluates the change in score associated with a candidate graph modification:

$$\Delta S = S(G_{\text{new}}; \mathbf{X}) - S(G_{\text{current}}; \mathbf{X})$$

A modification is accepted when it improves the score according to the search procedure.

The final result is generally a partially directed graph representing a class of observationally equivalent causal structures. Some directions may remain unresolved because the available observational data cannot distinguish between them.

The method is most naturally applied to independent observations from a stable distribution. For time series, temporal dependence must be modeled explicitly or the data must be transformed into an appropriate representation before applying a suitable implementation.

7. Causal Root Cause Analysis Using Directed Acyclic Graphs and Graph Neural Networks

Causal Root Cause Analysis using Directed Acyclic Graphs and Graph Neural Networks combines causal structure learning with anomaly localization. It uses a graph learning framework and a deep generative model to estimate relationships between monitored variables, including potentially nonlinear dependencies. The resulting graph can help identify variables that may explain an observed abnormality in an operational system.

A simplified linear structural equation is

$$\mathbf{X} = \mathbf{A}^T\mathbf{X} + \mathbf{Z}$$

where $\mathbf{X}$ represents observed variables across samples, $\mathbf{A}$ is the weighted adjacency matrix, and $\mathbf{Z}$ represents disturbances. Under this convention, the entry A_{ij} represents a candidate edge from X_i to X_j.

For the linear model, the equation can be rearranged as

$$\mathbf{X} = (\mathbf{I} - \mathbf{A}^T)^{-1}\mathbf{Z}$$

provided the inverse exists. In the nonlinear extension, the generative relationship can be represented more generally as

$$\mathbf{X} = f_\theta(\mathbf{A}, \mathbf{Z})$$

where f_θ is a learned function parameterized by θ. A graph neural network models how information flows between connected variables, while a generative model learns to reconstruct or explain the observed data.

For a directed acyclic graph formulation, a differentiable acyclicity constraint can be expressed as

$$h(\mathbf{A}) = \operatorname{tr}\left[(\mathbf{I} + \alpha\mathbf{A} \circ \mathbf{A})^d\right] - d = 0$$

where d is the number of variables and α is a positive scalar. This is one possible polynomial acyclicity constraint, rather than a universal objective for all graph neural network methods.

The model optimizes its graph and generative parameters to explain the observations while satisfying the chosen structural constraints. The learned graph is then used to assess how abnormalities may propagate through the system and to rank potential root causes.

A high root cause ranking does not by itself prove that a variable is the true causal origin. The reliability of the result depends on the quality of the learned graph, the anomaly detection method, the data, and the assumptions used to construct the model.

8. Comparison of the algorithms

Algorithm	Primary basis for causal discovery	Typical graph output	Supports cycles?
Linear Non Gaussian Acyclic Model	Independent non Gaussian disturbances and linear structural equations	Directed acyclic graph	No
Cyclic Linear Non Gaussian Model	Independent non Gaussian disturbances in a feedback system	Potentially cyclic directed graph	Yes, subject to additional assumptions
Granger causality	Improvement in prediction from historical source values	Directed lagged relationships	Yes, through lagged feedback
Peter and Clark	Conditional independence tests	Partially directed acyclic graph	No, in its standard formulation
Peter and Clark Momentary Conditional Independence	Conditional independence tests involving lagged variables	Time series causal graph	Supports lagged feedback
Noncombinatorial Optimization via Trace Exponential	Prediction error, sparsity penalty, and acyclicity constraint	Weighted directed acyclic graph	No, in its standard formulation
Greedy Equivalence Search	Graph score improvement and greedy search	Partially directed acyclic graph	No, in its standard formulation
Causal Root Cause Analysis using Graph Neural Networks	Generative model optimization and graph structure constraints	Weighted graph, often a directed acyclic graph	Depends on the specific formulation

9. Important considerations for implementation

The mathematical formulations above describe the principal mechanisms used by these algorithms. They do not imply that every nonzero edge is statistically significant or that every discovered relationship represents true causation.

The following considerations are particularly important when applying these methods to real world time series.

1. **Temporal dependence:** Standard versions of the Peter and Clark algorithm, Greedy Equivalence Search, Linear Non Gaussian Acyclic Model, and Noncombinatorial Optimization via Trace Exponential are not automatically designed to account for temporal dependence. Time series adaptations or suitable temporal modeling are required.

2. **Unobserved confounding:** Several methods rely on assumptions that relevant common causes have been observed. When this assumption is violated, inferred directions may be incorrect or remain unidentified.

3. **Statistical significance:** Conditional independence tests, predictive model comparisons, and optimized edge weights provide different kinds of evidence. Statistical testing, multiple testing correction, and stability analysis should be applied as appropriate.

4. **Graph interpretation:** Some algorithms identify a unique graph under strong assumptions, while others identify a class of graphs that cannot be distinguished using the available observational data alone.

5. **Cyclic relationships:** Feedback occurring across different time points can be modeled by temporal methods. Instantaneous cycles require a method that explicitly supports cyclic structural relationships.

The final algorithm selection should therefore depend on whether the data are temporal, whether instantaneous feedback is possible, whether nonlinear relationships are expected, and whether the objective is causal graph discovery, predictive analysis, or root cause localization.